

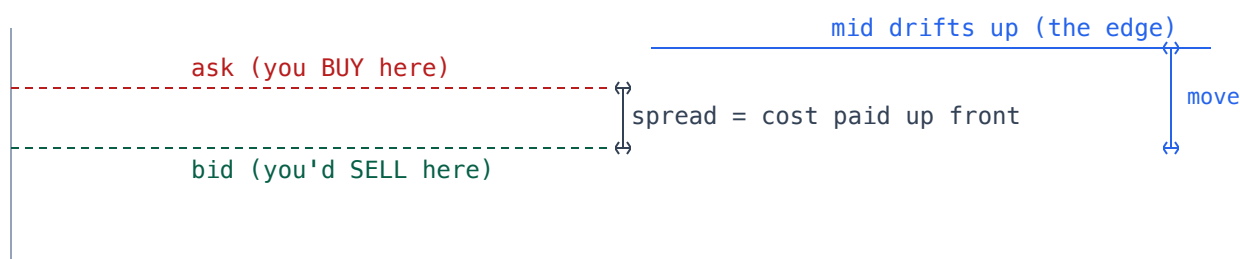
Does the signal make money? An honest P&L backtest

Learning notes for `src/backtest.cpp`. The signal bench proved we can go from book to order in tens of nanoseconds. Speed is necessary, not sufficient - the real question is whether acting on order-book imbalance **pays after costs**. This is the backtest, and its most valuable output is a *negative-sounding* result stated honestly, which is exactly what a research interviewer wants to see you reason about.

1. The one hard truth: signal vs spread

Every taker (a strategy that crosses the spread to trade *now*) pays the same tax: you buy at the **ask** and later sell at the **bid**, so a round trip forfeits roughly the **full spread** before the market moves at all. A directional signal only makes money if the move it predicts is **bigger than that spread**.

A taker round trip: pay the spread first, hope the move covers it



Profit needs $(\text{move}) > (\text{spread})$. OBI's move is small $\rightarrow$ that is the whole fight.

2. Building a flow where the signal *could* work

A backtest is only meaningful if the data has the effect you're testing for. We synthesize the flow from a **latent fair value** `FV` that random-walks. Order arrivals lean toward it: when the book is underpriced ($\text{FV} > \text{mid}$), buyers dominate and aggressive buys **eat the offer** - which does two things at once:

- lifts the **mid** up toward `FV` (the price move), and
- leaves the book **bid-heavy** (the imbalance).

So the imbalance genuinely *leads* the price - the real economic reason OBI works - rather than a circular `mid = alpha * OBI` identity baked in by hand. **This is a simulator, not the market:** its imbalance-to-price link is far cleaner than reality, which is why the accuracy below looks almost perfect. The machinery is what transfers; the numbers must be re-earned on live data.

3. Method: separate the market from the strategy

Phase 1 runs the sim once and records `{bid, ask, obi}` at every step. **Phase 2** replays that tape for several thresholds `T`. On `|OBI| > T` we open one unit crossing the spread and exit `H = 20` steps later crossing back. For each signal we log:

- **directional accuracy** - of the signals whose mid actually moved, the share that moved *our* way;
 - **gross mid-move** - the mid change we predicted, in ticks (the raw edge, before costs);
 - **net P&L / trade** - after crossing the spread both ways;
 - a **random-direction control** on the *same* trigger times - if OBI has real edge, it must beat this coin flip.
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4. What it found

thr	trades	dir-acc	grossMid	spread	net/trade	ctrl/trade	edge
0.20	69697	99.8%	+0.097	1.014	-0.917	-1.014	+0.097
0.30	48363	99.8%	+0.135	1.020	-0.885	-1.020	+0.135
0.40	15279	99.9%	+0.315	1.040	-0.725	-1.038	+0.313
0.50	461	99.7%	+0.818	1.087	-0.269	-0.996	+0.727
0.60	8	100.0%	+1.750	1.250	+0.500	-2.000	+2.500

Read it left to right as conviction rises:

1. **The signal is real.** Directional accuracy sits far above 50%, and every threshold beats the random control (positive `edge`). Higher conviction → bigger predicted move (`grossMid` climbs +0.10 → +1.75 ticks). That monotonic curve is the signature of a genuine signal.
 2. **But it loses as a taker** everywhere with a meaningful sample. `net/trade` is negative until `T ≈ 0.55`, because `grossMid < spread`: the move is real but smaller than the ~1-tick toll.
 3. **Profit only appears at the extremes**, where `grossMid` finally exceeds the spread - but only 8 trades qualify, far too few to trust.
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5. The conclusion worth stating

Order-book imbalance is a **directional** signal, not a **tradeable-by-crossing** one. The move it predicts is smaller than the spread, so a taker pays more to enter than the edge is worth.

The edge is real, so where does the money live? Two honest answers:

- **Provide, don't take.** Post a limit order on the favored side and *earn* the spread instead of paying it. Now imbalance tells you which side to quote - the sign flips from cost to income.
- **Use it as a filter/tilt**, not a standalone trade: skew an existing maker's quotes, or gate a different taker signal, so you never pay the spread just for the imbalance.

That is the real lesson of microstructure alpha: a true signal and a *profitable* one are different claims, separated by the spread. Being able to show that gap - with a control, a threshold sweep, and a stated caveat about the simulator - is worth more than a cherry-picked green P&L.

6. What would make this live-grade

- Replay **recorded L2 data** (the repo already records Binance.US depth) instead of a simulator, and expect accuracy to fall toward ~55%.
- Add a **maker/queue model** to test the "provide, don't take" version that this result points to.
- Report P&L in **currency with fees**, add inventory limits and a decay/half-life study of the signal, and bootstrap confidence intervals on `net/trade` .